

Tutorial 3 - Econometric Theory

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Triple Diff

Why one might want to use Triple Diff?

- Exploit Treatment Heterogeneity
- Remove PTA violation bias

Let's see theory first

Triple Diff

Say that:

$G \in \{0, 1\} \rightarrow$ Treated or Control groups (1)

$T \in \{0, 1\} \rightarrow$ Pre-post treatment time (2)

$H \in \{0, 1\} \rightarrow$ Sub-Groups differentially exposed to treatment (3)

where H is a *pre-treatment characteristic*

Triple Diff

DDD estimator is

$$\hat{\theta}_{DDD} = \hat{\delta}_{DD}^{H=1} - \hat{\delta}_{DD}^{H=0}$$

where

$$\hat{\delta}_{DD}^{H=1} = (\bar{y}_{11}^1 - \bar{y}_{10}^1) - (\bar{y}_{01}^1 - \bar{y}_{00}^1)$$

$$\hat{\delta}_{DD}^{H=0} = (\bar{y}_{11}^0 - \bar{y}_{10}^0) - (\bar{y}_{01}^0 - \bar{y}_{00}^0)$$

In principle, you are comparing two different DiDs (three differences, across TIME, GROUPS and SUBGROUPS)

Fully Saturated Regression

$$\begin{aligned} y_{cit} = & \beta_0 + \beta_1 G_i + \beta_2 Post_t + \beta_3 H_c \\ & + \beta_4(G_i \times Post_t) + \beta_5(G_i \times H_c) + \beta_6(Post_t \times H_c) \\ & + \theta(G_i \times Post_t \times H_c) + \epsilon_{cit} \end{aligned}$$

Three-way FE

$$\begin{aligned} y_{cit} = & \alpha_i + \delta_t + \gamma_c + \beta_4(G_i \times Post_t) + \beta_5(G_i \times H_c) + \beta_6(Post_t \times H_c) \\ & + \theta(G_i \times Post_t \times H_c) + \epsilon_{cit} \end{aligned}$$

Pair-wise FE Regression

$$y_{cit} = \phi_{ic} + \rho_{it} + \psi_{ct} + \theta(G_i \times Post_t \times H_c) + \epsilon_{cit}$$

Benefits and Costs of Triple Diff (cleaning bias case)

DDD cleans up two additional confounders, compared with basic DD:

- Group-specific (G) time trends common to both sub-groups H
- Sub-Group-Specific (H) time trends common to both groups G

Cost??? New Assumption → **Parallel Bias Assumption**

$$\begin{aligned} E[y(0)_{11} - y(0)_{10} | G = 1, H = 1] - E[y(0)_{11} - y(0)_{10} | G = 1, H = 0] \\ = \\ E[y(0)_{01} - y(0)_{00} | G = 0, H = 1] - E[y(0)_{01} - y(0)_{00} | G = 0, H = 0] \end{aligned}$$

In words: *In absence of the treatment, the differential trend between the $H=1$ and $H=0$ subgroups would have been the same in the treated group ($G=1$) as in the control group ($G=0$)*

Dataset(s) main characteristics

- 10000 units
- 40% treated
- time span = [2005,2020]
- First treatment date = 2012
- Two dummy covariates: X1 and X2

Case 1 - Bias Correction - How to interpret?

```
# A tibble: 3 × 3
  Estimator      Estimate StdError
  <chr>          <dbl>    <dbl>
1 Biased DiD, H = 1    10.6     0.162
2 Placebo DiD, H = 0    2.94     0.155
3 DDD                7.66     0.072
```

Interpretation: by assumption, both DiD are biased but share the same biased trend. If group $H=0$ is not affected by the treatment, then the placebo DiD is just a measure of the bias. So the true ATT is the difference between the biased DiD and the placebo one.

Example: imagine a treatment at the firm level, which is supposed to increase productivity. The treatment affect only women, but women productivity increaseing already more than male one (PTA violation). But, say this treatment affect only workers $< 30yo$. Then, one can use DiD estimate for $> 30yo$ to correct for the bias.

Case 2 – Heterogeneity - How to Interpret?

	<i>DDD heterogeneity</i>
<i>D</i>	6.595 (0.158)
<i>x1_post</i>	0.009 (0.038)
<i>hetero_term</i>	3.451 (0.061)
Observations	160,000
R^2	0.955
Unit FE	Yes
Year FE	Yes

Standard errors in parentheses.

Interpretation

- *D* captures the baseline treatment effect.
- *hetero_term* measures how the treatment effect varies between groups. In this case the coefficient is positive, hence ATT is higher for the H=1 group, compared to the H=0 one.

Example

Same example of previous slide. But now, the ATT affect productivity of all workers. But it affect differently Large and Small firms.

SUTVA and NA

There are 2 other assumptions needed before PTA, but most of the time go overlooked:

- **NA** (No Anticipation) → "no unit is effectively treated before it is formally treated"
- **SUTVA** (Stable Unit Treatment Value Assumption)
 - **No Interference** → "a treated unit cannot affect the potential outcomes of control units"
 - **No Hidden Variations of Treatment** → "all units assigned to the same treatment receive the same dose"

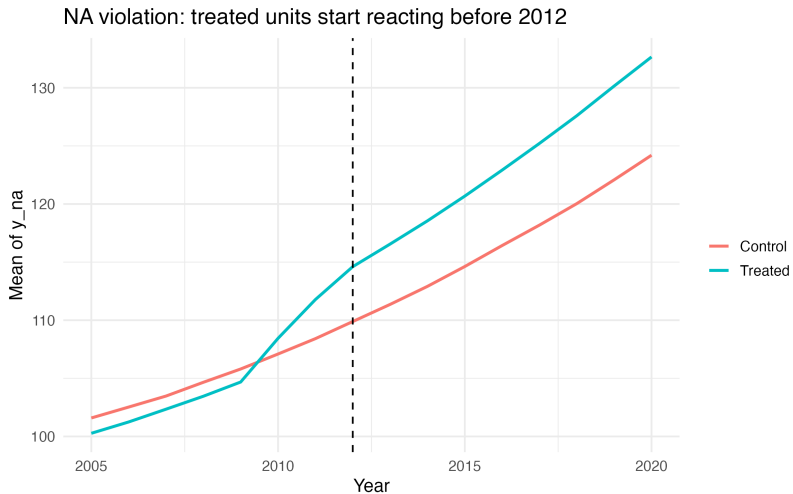
Let's see some examples to understand how they work

NA in potential outcome framework:

$$Y_{it}(0) = Y_{it} \forall t \leq T_0$$

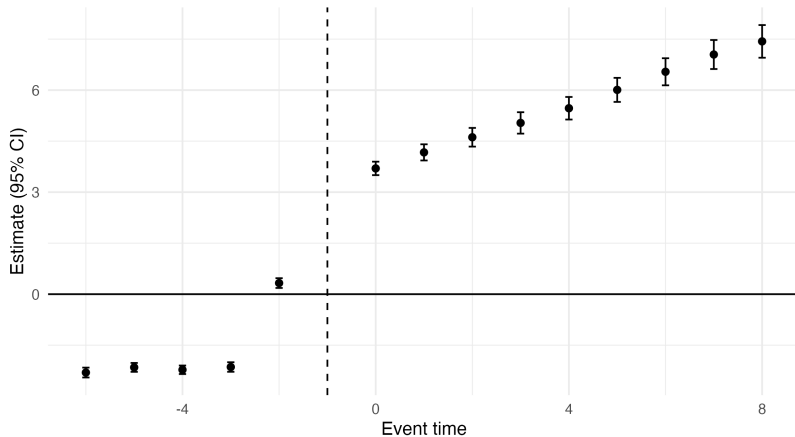
If violated:

$$\hat{\theta}_{DiD} = ATT_{Post} + \text{PTA-violation Bias} - \underbrace{ATT_{Pre}}_{\text{NA violation}}$$



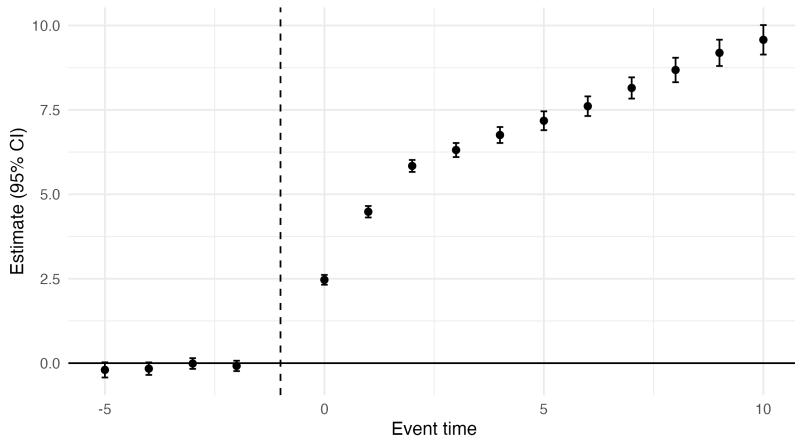
NA violation: event-study reveals significant leads

The significant leads are informative because the 'pre' period is already contaminated



Announcement-based design

If the announcement is the true treatment event, the design becomes coherent again



	<i>Naive DiD with NA violation</i>	<i>Re-dated announcement DiD</i>
<i>D</i>	6.770 (0.154)	
<i>announced</i>		7.022 (0.154)
Observations	160,000	160,000
R^2	0.951	0.950
Unit FE	Yes	Yes
Year FE	Yes	Yes

Standard errors in parentheses.

SUTVA I

Formally:

$$Y_i(W_i) \perp W_i \forall i \neq j$$

If violated, you will have:

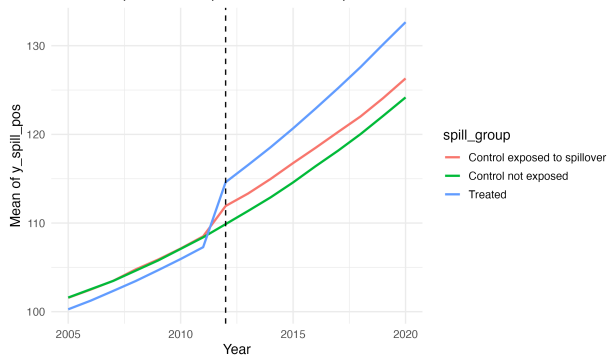
$$\hat{\theta}_{DiD} = ATT + \underbrace{\text{Spillover Bias}}_{\text{ATT on control group}}$$

Remedies:

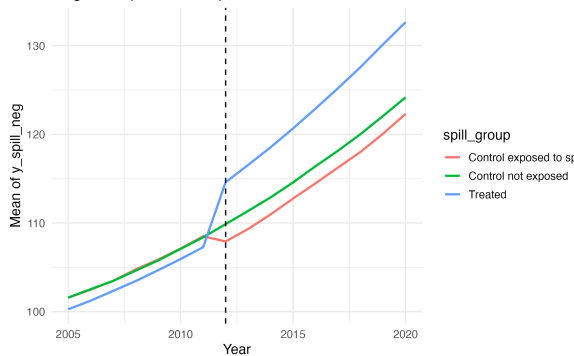
- Use geographically distant units as controls
- Explicitly model spillovers
- "Doughnut Design"

SUTVA I

Positive spillovers: exposed controls improve after treatment



Negative spillovers: exposed controls worsen after treatment

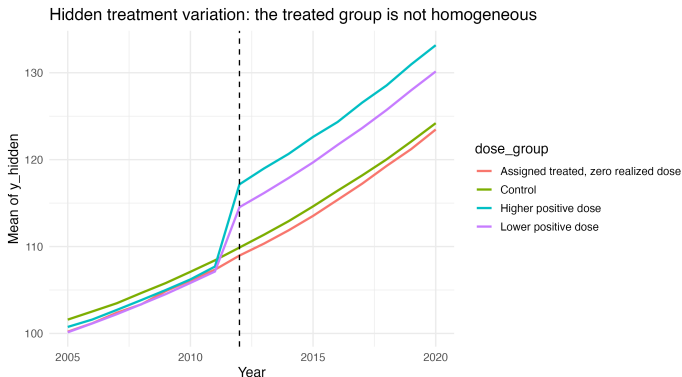


SUTVA I - Remedies

	Positive spillovers all controls	Positive spillovers doughnut design	Positive spillovers explicit exposure term
D	7.370 (0.171)	7.774 (0.162)	7.774 (0.162)
$I(\text{spill_exposed} \times \text{post})$			2.016 (0.266)
Observations	160,000	140,800	160,000
R^2	0.954	0.955	0.954
Unit FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes

Standard errors in parentheses.

SUTVA II



Where is the problem here? → You are estimating an average of different ATTs, so you cannot really say what you are estimating.

Synthetic Control

I would like to skip this part. If you are interested we will cover it in the next tutorial (text me). Just a few personal suggestions. :)

- Use synthetic control (all versions) only if necessary
- Preferred for small number of treated units (e.g. 1 treated unit)
- Reviewers don't like it. Use it as robustness check if not necessary
- Super easy to p-hack. All versions of it, just need to choose different covariates and you can get whatever you want.
- Apparently matching DiD is preferred by reviewers to SDiD.

Staggered Treatment - What you need to know?

From now on **I will borrow from Scott Cunningham lectures on staggered treatment.** You might see a lot of screenshots, but I think they are a cool and complete review of staggered treatment cases.

(I also borrow his notation...)

Bacon Decomposition

Consider TWFE:

$$Y_{ist} = \beta_0 + \delta D_{ist} + \tau_t + \sigma_s + \epsilon_{ist}$$

Say you have **differential treatment timing**. What is $\hat{\delta}$ estimating?

→ weighted average of all possible "four averages, three subtractions"

→ like doing a 2X2, when it is not. *Not great.*

Bacon Decomposition

Let's look at 3 timing groups (a, b and c) and one untreated group (U).
With 3 timing groups, there are 9 2x2 DDs. Here they are:

a to b	b to a	c to a
a to c	b to c	c to b
a to U	b to U	c to U

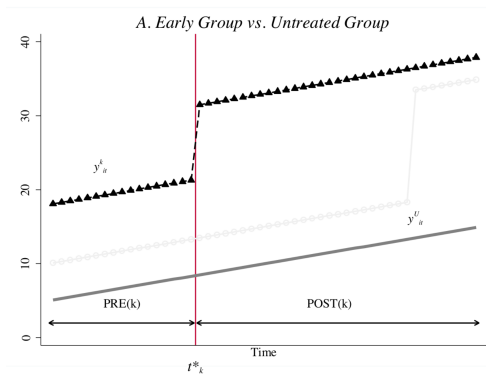
Let's return to a simpler example with only two groups – a k group treated at t_k^* and an l treated at t_l^* plus an never-treated group called the U untreated group

Bacon Decomposition

- Let there be two treatment groups (k, l) and one untreated group (U)
- k, l define the groups based on when they receive treatment (differently in time) with k receiving it earlier than l
- Denote $\overline{D}_k$ as the share of time each group spends in treatment status
- Denote $\widehat{\delta}_{jb}^{2 \times 2}$ as the canonical 2×2 DD estimator for groups j and b where j is the treatment group and b is the comparison group

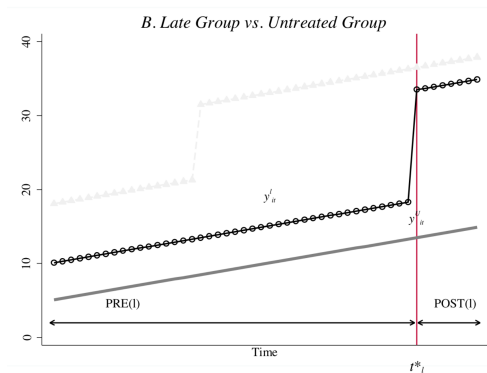
Bacon Decomposition

$$\hat{\delta}_{kU}^{2x2} = \left(\bar{y}_k^{post(k)} - \bar{y}_k^{pre(k)} \right) - \left(\bar{y}_U^{post(k)} - \bar{y}_U^{pre(k)} \right)$$



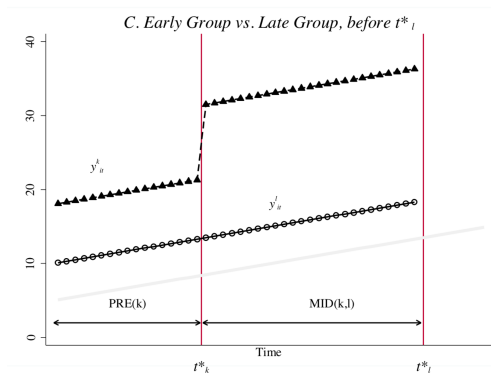
Bacon Decomposition

$$\widehat{\delta}_{lU}^{2x2} = \left(\bar{y}_l^{post(l)} - \bar{y}_l^{pre(l)} \right) - \left(\bar{y}_U^{post(l)} - \bar{y}_U^{pre(l)} \right)$$



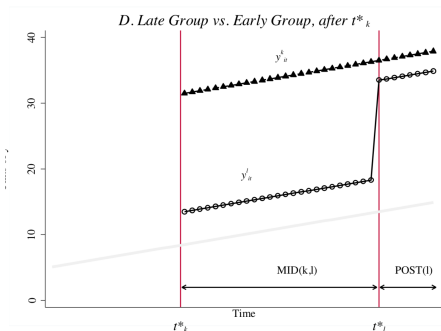
Bacon Decomposition

$$\delta_{kl}^{2x2,k} = \left(\bar{y}_k^{MID(k,l)} - \bar{y}_k^{Pre(k,l)} \right) - \left(\bar{y}_l^{MID(k,l)} - \bar{y}_l^{PRE(k,l)} \right)$$



Bacon Decomposition

$$\delta_{lk}^{2x2,l} = \left(\bar{y}_l^{POST(k,l)} - \bar{y}_l^{MID(k,l)} \right) - \left(\bar{y}_k^{POST(k,l)} - \bar{y}_k^{MID(k,l)} \right)$$



Bacon Decomposition

$$Y_{ist} = \beta_0 + \delta D_{ist} + \tau_t + \sigma_s + \varepsilon_{ist}$$

TWFE estimate of $\hat{\delta}$ is equal to a weighted average over all group 2x2 (of which there are 4 in this example)

$$\hat{\delta}^{TWFE} = \sum_{k \neq U} s_{kU} \hat{\delta}_{kU}^{2x2} + \sum_{k \neq U} \sum_{l > k} s_{kl} \left[\mu_{kl} \hat{\delta}_{kl}^{2x2,k} + (1 - \mu_{kl}) \hat{\delta}_{lk}^{2x2,l} \right]$$

where that first 2x2 combines the k compared to U and the l to U (combined to make the equation shorter)

Bacon Decomposition

$$\begin{aligned}s_{ku} &= \frac{n_k n_u \bar{D}_k (1 - \bar{D}_k)}{\widehat{Var}(\tilde{D}_{it})} \\ s_{kl} &= \frac{n_k n_l (\bar{D}_k - \bar{D}_l) (1 - (\bar{D}_k - \bar{D}_l))}{\widehat{Var}(\tilde{D}_{it})} \\ \mu_{kl} &= \frac{1 - \bar{D}_k}{1 - (\bar{D}_k - \bar{D}_l)}\end{aligned}$$

where n refer to the panel group shares, $\bar{D}_k(1 - \bar{D}_k)$, as well as $(\bar{D}_k - \bar{D}_l)(1 - (\bar{D}_k - \bar{D}_l))$ expressions refer to variance of treatment, and the final equation is the same for two timing groups.

Bacon Decomposition

- Two things to note:
 - More units in a group, the bigger its 2x2 weight is
 - Group treatment variance weights up or down a group's 2x2
- Think about what causes the treatment variance to be as big as possible. Let's think about the s_{ku} weights.
 - $\bar{D} = 0.1$. Then $0.1 \times 0.9 = 0.09$
 - $\bar{D} = 0.4$. Then $0.4 \times 0.6 = 0.24$
 - $\bar{D} = 0.5$. Then $0.5 \times 0.5 = 0.25$
 - $\bar{D} = 0.6$. Then $0.6 \times 0.4 = 0.24$
- This means the weight on treatment variance is maximized for *groups treated in middle of the panel*

Bacon Decomposition

- But what about the “treated on treated” weights (i.e., $\overline{D}_k - \overline{D}_l$)
- Same principle as before - when the difference between treatment variance is close to 0.5, those 2x2s are given the greatest weight
- For instance, say $t_k^* = 0.15$ and $t_l^* = 0.67$. Then $\overline{D}_k - \overline{D}_l = 0.52$. And thus $0.52 \times 0.48 = 0.2496$.

Bacon Decomposition

The Bacon decomposition shows that TWFE estimate of δ is unbiased under 2 assumptions:

- **Variance Weighted Parallel Trends are zero**
- **No Dynamic Treatment Effects**

Under these assumptions, TWFE estimators estimates the variance weighted ATT as a weighted average of all possible ATTs

Bacon Decomposition

$$\widehat{\delta}_{kU}^{2x2} = ATT_k Post + \Delta Y_k^0(Post(k), Pre(k)) - \Delta Y_U^0(Post(k), Pre)$$

$$\widehat{\delta}_{kl}^{2x2} = ATT_k(MID) + \Delta Y_k^0(MID, Pre) - \Delta Y_l^0(MID, Pre)$$

$$\begin{aligned} \widehat{\delta}_{lk}^{2x2} = & ATT_{l, Post(l)} + \underbrace{\Delta Y_l^0(Post(l), MID) - \Delta Y_k^0(Post(l), MID)}_{\text{Parallel trends bias}} \\ & - \underbrace{(ATT_k(Post) - ATT_k(Mid))}_{\text{Heterogeneity bias!}} \end{aligned}$$

Next

Enough for today.

Next time we will cover Callaway and Sant'Anna (2021) (and possibly some other estimators for staggered treatment, if you are interested).

Plus, tell me whether to do a review or not!!!